🧪 CHEMISTRY CH.1 — STRUCTURE OF ATOM — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this chapter:

1. Subatomic Particles & Atomic Structure Basics

- Proton = positively charged, nucleus mein hota hai; mass ≈ 1 u; charge = +e. *[Q3, Q5, Q7]*
- Neutron = neutral (no charge), nucleus mein hota hai; mass ≈ 1 u. *[Q3, Q5, Q7]*
- Electron = negatively charged, nucleus ke charon ore orbit/orbital mein; mass ≈ 1/1836 u (negligible); charge = −e. *[Q3, Q5, Q7]*
- Nucleus mein sirf protons + neutrons hain; electrons bahar revolve karte hain. *[Q3, Q5]*
- Neutrons nikaalne ka formula: n = A − Z. *[Q6, Q8, Q71]*

2. Atomic Number, Mass Number & Isotopes

- Atomic Number (Z) = number of protons = number of electrons (neutral atom mein). *[Q4, Q6, Q8, Q71, Q77]*
- Mass Number (A) = protons + neutrons. *[Q4, Q6, Q8, Q71, Q77]*
- Isotopes = same atomic number, different mass number; jaise C-12, C-13, C-14. *[Q9, Q11]*

- Isobars = same mass number, different atomic number; jaise Ar-40, Ca-40. *[Q9, Q11]*
- Isotones = same number of neutrons; jaise C-14 (8n), N-15 (8n). *[Q9, Q11]*
- Average atomic mass = Σ(isotope mass × relative abundance); example: Cl-35 (75%) + Cl-37 (25%) = 35.5 u. *[Q77]*

3. Atomic Mass, Molecular Mass & Mole Concept

- Atomic Mass Unit (amu/u) = 1/12th of C-12 atom ka mass; 1 u ≈ 1.66×10^{-27} kg. *[Q2, Q25]*
- Carbon = 12 u; Hydrogen = 1 u; Oxygen = 16 u; Nitrogen = 14 u. *[Q2]*
- Molecular mass = atoms ke atomic masses ka sum; H_2O = 18 u; CO_2 = 44 u. *[Q25, Q27]*
- Mole concept: 1 mole = 6.022×10^{23} particles (Avogadro's number). *[Q27, Q29]*
- 1 mole ka mass (grams mein) = molecular mass (u mein); 1 mole H_2O = 18 g. *[Q27]*
- Molecules & Atomicity = ek molecule mein atoms ki sankhya: Monoatomic — Ar, Ne (1 atom); Diatomic — H_2, N_2, O_2, Cl_2 (2 atoms); Tetratomic — P_4 (4 atoms); Octatomic — S_8 (8 atoms, common elements mein sabse zyada). *[Q1, Q26, Q28]*
- Compound = do ya zyada different elements ka fixed ratio mein chemical bonding se bana substance; Water (H_2O), Ammonia (NH_3), Nitrogen peroxide sab compounds hain, lekin Chlorine ek element hai (compound nahi). *[Q32]*
- Molecule = compound ka smallest possible unit — do ya zyada atoms ka definite arrangement mein aggregate; Atom = element ka smallest particle; Nucleus = atom ke center ka positively charged region. *[Q58]*

4. Discovery Timeline & Scientists

- Dalton (1808): Atomic theory — indivisible atoms. *[Q62, Q64]*
- J.J. Thomson (1897): Electron discover; cathode ray experiment; plum pudding model; e/m ratio measure ki. *[Q38, Q40, Q62, Q64]*
- Millikan: Oil-drop experiment — electron ka exact charge ($e = 1.6 \times 10^{-19}$ C). *[Q42]*
- Goldstein: Canal rays / positive rays discover ki — protons ka indication. *[Q66]*
- Ernest Rutherford (1911): Gold foil experiment; nucleus discover; proton discover (1919); planetary model. *[Q38, Q39, Q40, Q41, Q62, Q64]*
- James Chadwick (1932): Neutron discover; beryllium par alpha particle bombard karke. *[Q38, Q40, Q62, Q64]*
- Niels Bohr (1913): Quantized orbits; hydrogen spectrum explain ki. *[Q22, Q44, Q62, Q64]*
- Schrödinger (1926): Wave equation; quantum mechanical model. *[Q68]*
- Heisenberg: Uncertainty Principle — position aur momentum simultaneously exact nahi. *[Q68, Q70]*
- Moseley: Atomic number ka concept diya X-ray studies se. *[Q59, Q66]*

5. Atomic Models: Thomson → Rutherford → Bohr → Quantum

- Thomson's Plum Pudding Model: Positive sphere mein negative electrons embedded; uniform positive charge. *[Q13, Q15, Q61, Q63]*
- Rutherford's Nuclear Model (Gold Foil Experiment): Atom ka most part khali hai; positive charge small dense nucleus mein concentrated; nucleus ka size atom se bahut chhota (≈10^{-15} m vs atom ≈10^{-10} m); electrons nucleus ke charon ore revolve karte hain. *[Q13, Q15, Q39, Q41, Q43]*
- Rutherford model ka drawback: classical physics ke hisaab se revolving electron continuously energy radiate karke spiral mein nucleus mein gir jaana chahiye — ye nahi hota. *[Q20, Q43, Q63, Q65]*
- Bohr's Model: Electrons definite circular orbits/energy levels mein revolve karte hain bina energy lose kiye; har

orbit ki fixed energy hai = energy level/shell (K, L, M, N...); energy absorb/release karke electron shell change karta hai; angular momentum quantized hai: $mvr = nh/2\pi$. *[Q22, Q24, Q44, Q46, Q65, Q67]*

- Bohr model ka limitation: sirf hydrogen-like species (single electron) ke liye accurate; multi-electron atoms ke spectral lines explain nahi kar paata. *[Q48, Q67, Q69]*
- Quantum Mechanical Model (Schrödinger): Electron ko wave-particle duality ke saath describe kiya; exact location nahi, sirf probability — electron cloud/orbital; Heisenberg Uncertainty Principle: $\Delta x \times \Delta p \geq h/4\pi$. *[Q68, Q70, Q72]*

6. Quantum Numbers, Orbitals & Shapes

- Principal quantum number (n) = shell; values 1, 2, 3...; energy aur size decide karti hai. *[Q45, Q47]*
- Azimuthal/Angular momentum quantum number (l) = subshell; values 0 se (n−1) tak: l=0 → s; l=1 → p; l=2 → d; l=3 → f. *[Q45, Q47]*
- Magnetic quantum number (m_l) = orbital orientation; values −l se +l tak: s — 1 orbital; p — 3 orbitals; d — 5 orbitals; f — 7 orbitals. *[Q47]*
- Spin quantum number (m_s) = electron spin; values +½ ya −½. *[Q47]*
- Total electrons in a shell = $2n^2$; in a subshell = $2(2l+1)$. *[Q45]*
- Orbital shapes: s-orbital — spherical, 2 electrons max; p-orbital — dumb-bell, 6 electrons max; d-orbital — cloverleaf/double dumb-bell, 10 electrons max; f-orbital — complex, 14 electrons max. *[Q73, Q75]*
- Orbital = 3D space jahan electron milne ki probability max (≈90–95%) hoti hai; orbit (definite path) se alag hai. *[Q72, Q74]*
- Orbital size n ke saath badhti hai; energy bhi badhti hai. *[Q74]*

7. Electronic Configuration: Rules & Examples

- Aufbau Principle: Electrons pehle lowest energy orbital mein bharte hain. *[Q49, Q51]*
- Filling order: 1s → 2s → 2p → 3s → 3p → 4s → 3d → 4p → 5s → 4d → 5p → 6s → 4f → 5d → 6p... *[Q49]*
- Pauli Exclusion Principle: Ek orbital mein max 2 electrons ho sakte hain aur unke spins opposite (↑↓) hone chahiye. *[Q49, Q51, Q76, Q78]*
- Hund's Rule of Maximum Multiplicity: Ek subshell ke orbitals mein electrons pehle singly bharte hain (same spin), phir pairing hoti hai. *[Q49, Q51, Q76, Q78]*
- Common configurations: H (Z=1) — $1s^1$; He (Z=2) — $1s^2$; C (Z=6) — $1s^2\ 2s^2\ 2p^2$; N (Z=7) — $1s^2\ 2s^2\ 2p^3$ (↑ ↑ ↑); O (Z=8) — $1s^2\ 2s^2\ 2p^4$; Na (Z=11) — [Ne] $3s^1$; Ar (Z=18) — $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$; K (Z=19) — [Ar] $4s^1$. *[Q50, Q51, Q52, Q54]*
- Exceptions (stability ke liye): Cr (Z=24) — [Ar] $3d^5\ 4s^1$ (half-filled d-subshell stability); Cu (Z=29) — [Ar] $3d^{10}\ 4s^1$ (fully-filled d-subshell stability). *[Q56]*
- Ions ki configuration: Fe (Z=26) [Ar] $3d^6\ 4s^2$ → Fe^{2+}: [Ar] $3d^6$; Fe^{3+}: [Ar] $3d^5$. Cu (Z=29) [Ar] $3d^{10}\ 4s^1$ → Cu^+: [Ar] $3d^{10}$. Cl (Z=17) [Ne] $3s^2\ 3p^5$ → Cl^-: [Ne] $3s^2\ 3p^6$ (octet). *[Q79, Q81]*

8. Valence Electrons, Valency & Chemical Properties

- Shells K, L, M, N... ya 1, 2, 3, 4... se denote hoti hain. *[Q10, Q12, Q14]*
- Max electrons in a shell: $2n^2$; K=2, L=8, M=18, N=32. *[Q10, Q12]*
- Outermost shell = valence shell; uske electrons = valence electrons; chemical properties ye decide karte hain. *[Q14, Q16, Q18, Q53, Q55]*
- Outermost shell mein max 8 electrons ho sakte hain (octet rule). *[Q14, Q16]*
- Valency = atom ka combining capacity: valence electrons ≤ 4 → valency = valence electrons (donate karega);

valence electrons > 4 → valency = 8 − valence electrons (accept karega). *[Q17, Q19, Q21]*

- Group 1 (alkali metals): 1 valence electron — highly reactive. *[Q55]*
- Group 17 (halogens): 7 valence electrons — 1 electron accept karte hain. *[Q55]*
- Group 18 (noble gases): 8 valence electrons (except He: 2) — chemically inert. *[Q19, Q21, Q55]*
- Metals generally 1–3 valence electrons; electrons donate karte hain. Non-metals generally 5–7 valence electrons; electrons accept karte hain. *[Q21]*
- Covalently bonded molecules ka melting/boiling point low hota hai kyunki inke beech ki intermolecular forces weak hoti hain (covalent bond mein atoms electrons share karte hain). *[Q60]*

9. Ions: Cations & Anions

- Ion = charged atom/molecule; electron loss/gain se banta hai. *[Q23, Q30]*
- Cation = positive ion; electron donate karke banta hai; generally metals. $Na \rightarrow Na^+ + e^-$; $Mg \rightarrow Mg^{2+} + 2e^-$. *[Q23]*
- Anion = negative ion; electron accept karke banta hai; generally non-metals. $Cl + e^- \rightarrow Cl^-$; $O + 2e^- \rightarrow O^{2-}$. *[Q23]*
- Polyatomic ions: NH_4^+, OH^-, SO_4^{2-}, CO_3^{2-}, NO_3^-. *[Q30]*

10. Radioactivity, Nuclear Reactions & Applications

- Radioactivity = unstable nucleus ka spontaneous disintegration; alpha, beta, gamma rays emit hoti hain. *[Q31, Q33, Q35, Q80]*
- Alpha (α) = Helium nucleus ($_2He^4$); +2 charge; least penetrating, most ionizing; paper se roka ja sakta hai. *[Q31, Q33, Q80]*
- Beta (β) = high-speed electron ($_{-1}e^0$); −1 charge; moderate penetrating; aluminium se roka ja sakta hai. *[Q31, Q33, Q80]*
- Gamma (γ) = EM waves; no charge; most penetrating, least ionizing; thick lead/concrete se roka ja sakta hai. *[Q31, Q33, Q80]*
- Penetrating power: $\alpha < \beta < \gamma$; Ionizing power: $\alpha > \beta > \gamma$. *[Q31, Q33]*
- Half-life ($t_{1/2}$) = time in which half of radioactive atoms decay kar jaate hain. *[Q35, Q37]*
- Example: 1g → 0.5g → 0.25g in 12 days means 2 half-lives; $t_{1/2}$ = 6 days. *[Q37]*
- Nuclear fission = heavy nucleus (U-235, Pu-239) split; chain reaction; nuclear reactor/atomic bomb. *[Q34, Q36]*
- Nuclear fusion = light nuclei combine; sun/stars; hydrogen bomb; ~100 million °C chahiye. *[Q34, Q36]*
- Radioactive decay types: Alpha decay — mass −4, atomic number −2; Beta decay — mass same, atomic number +1; Gamma decay — mass & atomic number same. *[Q80]*
- Applications: C-14 dating (5730 years), Co-60 cancer treatment, I-131 thyroid diagnosis. *[Q80]*

11. Periodic Trends: Radius, IE, EA & Electronegativity

- Atomic radius = nucleus se outermost electron shell ki distance. Left to right (period): decrease (nuclear charge badhta hai); Top to bottom (group): increase (new shell add hoti hai). *[Q82, Q84]*
- Ionic radius: Cation < Parent atom (electron loss se effective nuclear charge badhta hai); Anion > Parent atom (electron gain se repulsion badhta hai). *[Q84]*
- Isoelectronic species mein: jyada nuclear charge → chhota radius. $N^{3-} > O^{2-} > F^- > Ne > Na^+ > Mg^{2+} > Al^{3+}$ (sabke paas 10 electrons). *[Q86]*
- Ionization Energy (IE) = gaseous atom se outermost electron remove karne ki energy. Left to right: increase; Top to bottom: decrease. *[Q83, Q85]*

- Electron Affinity (EA) = gaseous atom mein electron add karne par release hone wali energy. *[Q83]*
- Electronegativity = shared electron pair ko apni taraf khinchne ki tendency; Pauling scale par max Fluorine (F = 4.0). *[Q85]*
- Same period ke Si, Na, Mg, Al mein sabse bada ion Na banata hai — kam nuclear charge ki wajah se electron loss ke baad bhi ionic radius sabse zyada rehta hai. *[Q57]*

12. ⚠ Traps / Key Differences Summary (Q1–Q86)

- ⚠ Trap: Neon (Ne) monoatomic hai — diatomic nahi; question mein intentionally confuse kiya tha. *[Q1]*
- ⚠ Trap: Sulphur S_8 (8 atoms) — common elements mein sabse zyada atomicity. *[Q1]*
- ⚠ Trap: Isotopes = same Z, different A; Isobars = same A, different Z — ulta mat samajhna. *[Q9, Q11]*
- ⚠ Trap: Electron nucleus mein nahi hota; nucleus mein sirf protons + neutrons hain. *[Q3, Q5]*
- ⚠ Trap: Orbit = definite path (Bohr model); Orbital = probability cloud (quantum model) — donon alag hain. *[Q72, Q74]*
- ⚠ Trap: Penetrating power: $\alpha < \beta < \gamma$; Ionizing power: $\alpha > \beta > \gamma$ — reverse relation hai. *[Q31, Q33]*
- ⚠ Trap: Gamma rays electromagnetic waves hain, charged particles nahi — isliye sabse zyada penetrate karti hain. *[Q31]*
- ⚠ Trap: Half-life mein sample ka mass exponentially decrease hota hai, linear nahi. *[Q37]*
- ⚠ Trap: Rutherford ne nucleus/proton discover kiya; Chadwick ne neutron — donon alag hain. *[Q38, Q40]*
- ⚠ Trap: Bohr model sirf hydrogen-like atoms ke liye accurate hai; multi-electron atoms ke liye nahi. *[Q48, Q67, Q69]*
- ⚠ Trap: 4s orbital 3d se pehle bharata hai Aufbau principle ke hisaab se — K (Z=19) ka configuration [Ar] $4s^1$ hai, 3d nahi. *[Q54]*
- ⚠ Trap: Cr (Z=24) aur Cu (Z=29) exceptions hain — half-filled aur fully-filled d-subshell ki stability ke kaaran. *[Q56]*
- ⚠ Trap: Cation bante waqt electrons outermost shell se pehle nikle; Fe^{2+} mein 4s se electrons gaye, 3d se nahi. *[Q79]*
- ⚠ Trap: Fe^{3+} [Ar] $3d^5$ aur Cu^+ [Ar] $3d^{10}$ — half-filled aur fully-filled d-subshell ki extra stability. *[Q79, Q81]*
- ⚠ Trap: Isoelectronic species mein jyada nuclear charge wala chhota hota hai — $Na^+ > Mg^{2+} > Al^{3+}$ mein radius ghatta hai. *[Q86]*
- ⚠ Trap: IE, EA, Electronegativity — teenon periodic trends similar hain (left→right increase, top→bottom decrease) lekin exceptions hain (noble gases, Be, N). *[Q83, Q85]*
- ⚠ Trap: Fluorine (F) sabse zyada electronegative hai — Oxygen nahi. *[Q85]*
- ⚠ Trap: Average atomic mass always closest to most abundant isotope hoti hai. *[Q77]*
- ⚠ Trap: Moseley's Law se atomic number determine hota hai, atomic mass nahi. *[Q59]*
- ⚠ Trap: 1 mole = 6.022×10^{23} particles; yaad rakhna — "mole" number nahi, amount hai. *[Q27, Q29]*
- ⚠ Trap: Cation positive hai (electron loss), Anion negative hai (electron gain) — sign confuse mat hona. *[Q23]*
- ⚠ Trap: Chlorine ek element hai, compound nahi — compound banane ke liye do ya zyada different elements chahiye. *[Q32]*
- ⚠ Trap: Covalent molecules ki weak intermolecular forces ki wajah se low melting/boiling point hota hai. *[Q60]*
- ⚠ Trap: Same period mein sabse kam nuclear charge wale element (jaise Na) ka ion sabse bada hota hai. *[Q57]*

13. Numerical: Atomic Number, Mass Number & Subatomic Particles

- Formula: Atomic number (Z) = Protons = Mass number (A) − Neutrons. *[Q87, Q89, Q91]*

- A = 298, neutrons = 189 → Z = 298 − 189 = 109. *[Q87]*
- A = 40, neutrons = 20 → Z = 40 − 20 = 20 (Calcium). *[Q89]*
- Z = 17, neutrons = 18 → A = 17 + 18 = 35 (Chlorine-35). *[Q91]*

14. Numerical: Moles, Molar Mass & Mass Conversions

- Formula: Moles = Given mass (g) / Molar mass (g/mol). *[Q88, Q90, Q92, Q94, Q96]*
- H_2SO_4 ka molar mass = 2(1) + 32 + 4(16) = 98 g/mol; 25 g → 25/98 = 0.255 mol. *[Q88]*
- NaOH ka molar mass = 23 + 16 + 1 = 40 g/mol; 20 g → 20/40 = 0.5 mol. *[Q90]*
- $CaCO_3$ ka molar mass = 40 + 12 + 3(16) = 100 g/mol; 50 g → 50/100 = 0.5 mol. *[Q92]*
- H_2O ka molar mass = 18 g/mol; 9 g → 9/18 = 0.5 mol. *[Q94]*
- CO_2 ka molar mass = 12 + 2(16) = 44 g/mol; 22 g → 22/44 = 0.5 mol. *[Q96]*
- $Ca(OH)_2$ ka relative mass = Ca (40.08) + 2×O (31.998) + 2×H (2.016) = 74 u. *[Q107]*
- 90 g water mein moles = mass/molar mass = 90/18 = 5 moles. *[Q108]*
- CO (Carbon monoxide) ka relative mass = C (12) + O (16) = 28. *[Q109]*
- Ammonia (NH_3) mein N_2 aur H_2 ke masses ka ratio hamesha 14 : 3 hota hai (N ka mass 14, 3 H atoms ka mass 3). *[Q110]*
- 60 g Helium mein moles = mass/molar mass = 60/4 = 15 moles (He ka molar mass = 4 g). *[Q137]*

15. Numerical: Number of Atoms, Molecules & Ions from Moles

- Formula: Number of particles = Moles × Avogadro's number (6.022×10^{23}). *[Q93, Q95, Q97, Q99]*
- 0.5 mol H_2O mein molecules = $0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$. *[Q93]*
- 1 mol CO_2 mein molecules = 6.022×10^{23}. *[Q95]*
- 0.25 mol NaCl mein formula units = $0.25 \times 6.022 \times 10^{23} = 1.5055 \times 10^{23}$. *[Q97]*
- 2 mol O_2 mein molecules = $2 \times 6.022 \times 10^{23} = 1.2044 \times 10^{24}$. *[Q99]*

16. Numerical: Gram Atoms, Gram Molecules & Avogadro's Number

- 1 gram atom = 1 mole atoms = 6.022×10^{23} atoms; mass in grams = atomic mass. *[Q111, Q113, Q115]*
- 1 gram molecule = 1 mole molecules = 6.022×10^{23} molecules; mass in grams = molecular mass. *[Q111, Q113]*
- 1 gram atom oxygen = 16 g mein 6.022×10^{23} atoms. *[Q111]*
- 1 gram molecule H_2O = 18 g mein 6.022×10^{23} molecules. *[Q113]*
- 1 gram atom sodium = 23 g mein 6.022×10^{23} atoms. *[Q115]*

17. Numerical: Molar Mass from Vapour Density & Gas Laws

- Formula: Molar mass = 2 × Vapour density (VD). *[Q98, Q100, Q102]*
- VD = 14 → Molar mass = 2 × 14 = 28 g/mol (N_2 ya CO). *[Q98]*
- VD = 22 → Molar mass = 2 × 22 = 44 g/mol (CO_2). *[Q100]*
- VD = 32 → Molar mass = 2 × 32 = 64 g/mol (SO_2). *[Q102]*

18. Numerical: Number of Molecules in Given Volume at STP

- STP par: 1 mole gas = 22.4 L volume leti hai. *[Q122, Q124, Q126]*
- 22.4 L gas at STP = 1 mole = 6.022×10^{23} molecules. *[Q123, Q125, Q127]*
- 11.2 L gas at STP = 11.2/22.4 = 0.5 mole → 3.011×10^{23} molecules. *[Q123]*
- 5.6 L gas at STP = 5.6/22.4 = 0.25 mole → 1.5055×10^{23} molecules. *[Q125]*
- 2.24 L gas at STP = 2.24/22.4 = 0.1 mole → 6.022×10^{22} molecules. *[Q127]*
- 5.6 L gas ka mass 11 g → Molar mass = 11 × 22.4 / 5.6 = 44 g/mol. *[Q122]*
- 2.24 L gas ka mass 4.4 g → Molar mass = 4.4 × 22.4 / 2.24 = 44 g/mol. *[Q124]*
- 11.2 L gas ka mass 22 g → Molar mass = 22 × 22.4 / 11.2 = 44 g/mol. *[Q126]*

19. Numerical: Percentage Composition

- Formula: % of element = (Mass of element in compound / Molar mass of compound) × 100. *[Q101, Q103, Q105]*
- H_2O mein H% = (2/18) × 100 = 11.11%. *[Q101]*
- $CaCO_3$ mein Ca% = (40/100) × 100 = 40%. *[Q103]*
- NH_3 mein N% = (14/17) × 100 = 82.35% ≈ 82.4%. *[Q105]*

20. Numerical: Empirical & Molecular Formula

- Empirical formula = simplest whole number ratio of atoms. *[Q104, Q106]*
- Molecular formula = n × Empirical formula; jahan n = Molecular mass / Empirical formula mass. *[Q104]*
- Empirical formula CH_2, molecular mass = 42 → Empirical mass = 14 → n = 42/14 = 3 → Molecular formula = C_3H_6. *[Q104]*

21. Numerical: Number of Electrons, Protons & Neutrons in Ions

- Neutral atom mein: Protons = Electrons = Atomic number (Z). *[Q112, Q114, Q116]*
- Cation mein: Electrons = Z − positive charge; protons same. *[Q112, Q116]*
- Anion mein: Electrons = Z + negative charge; protons same. *[Q114]*
- Na^+ (Z=11): protons = 11, electrons = 10, neutrons = 23 − 11 = 12. *[Q112]*
- Cl^- (Z=17): protons = 17, electrons = 18, neutrons = 35 − 17 = 18. *[Q114]*
- Mg^{2+} (Z=12): protons = 12, electrons = 10, neutrons = 24 − 12 = 12. *[Q116]*
- Al^{3+} (Z=13): protons = 13, electrons = 10, neutrons = 27 − 13 = 14. *[Q118]*
- O^{2-} (Z=8): protons = 8, electrons = 10, neutrons = 16 − 8 = 8. *[Q120]*

22. Numerical: Isotopic Mass & Average Atomic Mass

- Average atomic mass = Σ(Isotope mass × Relative abundance) / 100. *[Q117, Q119, Q121]*
- Boron: B-10 (20%) + B-11 (80%) → Average = (10×20 + 11×80)/100 = 10.8 u. *[Q117]*
- Chlorine: Cl-35 (75%) + Cl-37 (25%) → Average = 35.5 u. *[Q119]*
- Neon: Ne-20 (90.51%), Ne-21 (0.27%), Ne-22 (9.22%) → Average ≈ 20.18 u. *[Q121]*

23. Numerical: Electron Configuration & Quantum Numbers

- Z = 11 (Na): $1s^2 2s^2 2p^6 3s^1$; valence electron = 1. *[Q128, Q130]*
- Z = 17 (Cl): $1s^2 2s^2 2p^6 3s^2 3p^5$; valence electron = 7. *[Q128, Q130]*
- Z = 20 (Ca): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$; valence electron = 2. *[Q132]*
- Z = 26 (Fe): [Ar] $3d^6 4s^2$; valence electrons = 2. *[Q134]*
- Z = 29 (Cu) exception: [Ar] $3d^{10} 4s^1$; valence electron = 1. *[Q134]*
- Last electron ki quantum numbers: Na — n=3, l=0, m_l=0, m_s=+½; Cl — n=3, l=1, m_l=+1, m_s=−½. *[Q133, Q135]*

24. Numerical: Maximum Electrons in Shells/Subshells

- Shell mein max electrons = $2n^2$. *[Q129, Q131]*
- K=2, L=8, M=18, N=32. *[Q129]*
- Subshell mein max: s=2, p=6, d=10, f=14. *[Q131]*

25. Numerical: Wavelength, Frequency & Energy of Electron Transitions

- Bohr energy: $E_n = -13.6\, Z^2/n^2$ eV. *[Q136, Q138, Q140]*
- Hydrogen ground state n=1: −13.6 eV. *[Q136]*
- $\Delta E = -13.6\,(1/n_2^2 - 1/n_1^2)$ eV. *[Q138]*
- n=2 → n=1: ΔE = 10.2 eV. *[Q138]*
- Rydberg formula: $1/\lambda = R\,(1/n_1^2 - 1/n_2^2)$. *[Q140]*
- Lyman series: n=1 (UV); Balmer: n=2 (visible); Paschen: n=3; Brackett: n=4. *[Q140]*

26. Numerical: Limiting Reagent & Stoichiometry

- Limiting reagent = reaction mein pehle khatam hone wala reactant. *[Q139, Q141]*
- $2H_2 + O_2 \rightarrow 2H_2O$; 4 g H_2 (2 mol) + 32 g O_2 (1 mol) → O_2 limiting; H_2O = 36 g. *[Q139]*
- $N_2 + 3H_2 \rightarrow 2NH_3$; 28 g N_2 (1 mol) + 6 g H_2 (3 mol) → exact ratio; NH_3 = 34 g. *[Q141]*

27. Numerical: Formula Cheat Sheet

- Atomic number: Z = A − n.
- Moles: n = mass / molar mass.
- Particles: $N = n \times N_a$ ($N_a = 6.022 \times 10^{23}$).
- Vapour density: M = 2 × VD.
- % Composition: (mass of element / molar mass) × 100.
- STP volume: 1 mol gas = 22.4 L.
- Molar mass from STP: M = mass × 22.4 / Volume.
- Bohr energy: $E_n = -13.6\, Z^2/n^2$ eV.

- Rydberg formula: $1/\lambda = R (1/n_1^2 - 1/n_2^2)$.
- Limiting reagent: Compare mole ratio with balanced equation.

28. ⚠️ Traps / Key Differences Summary (Numerical Q87–Q141)

- ⚠️ Trap: Atomic number = Mass number − Neutrons; electrons isme include nahi hote. *[Q87]*
- ⚠️ Trap: Molar mass calculate karte waqt atomic masses ka accurate sum karo; H=1, O=16, C=12, N=14, S=32, Na=23, Ca=40, Cl=35.5. *[Q88, Q90]*
- ⚠️ Trap: Vapour density ka unit nahi hota; ye hydrogen gas ke density se ratio hai. *[Q98]*
- ⚠️ Trap: Empirical formula mass aur molecular mass alag hain; n nikaalna mat bhoolna. *[Q104]*
- ⚠️ Trap: Ion mein protons aur neutrons same rehte hain; sirf electrons ki sankhya badalti hai. *[Q112, Q114, Q116]*
- ⚠️ Trap: Na^+, Mg^{2+}, Al^{3+}, O^{2-} sabke paas 10 electrons hain (isoelectronic) lekin protons alag hain. *[Q112, Q116, Q118, Q120]*
- ⚠️ Trap: Average atomic mass mein percentage abundance se multiply karna hota hai; simple average nahi. *[Q117, Q119]*
- ⚠️ Trap: STP = 0°C, 1 atm; iske alag conditions par 22.4 L formula galat hoga. *[Q122]*
- ⚠️ Trap: Gram atom aur Gram molecule ab "mole" ke terms mein samajhe jaate hain. *[Q111, Q113]*
- ⚠️ Trap: Quantum numbers nikaalne mein last electron ka spin dhyan se dekho — Hund's rule se pairing pattern samajhna zaroori hai. *[Q135]*
- ⚠️ Trap: Lyman series UV mein hai; Balmer visible mein — ye regions yaad rakhna exam mein direct puchha jaata hai. *[Q140]*
- ⚠️ Trap: Limiting reagent identify karne ke liye balanced equation ka mole ratio compare karo; mass directly compare mat karo. *[Q139, Q141]*
- ⚠️ Trap: Bohr energy formula sirf hydrogen-like species (single electron) ke liye exact hai. *[Q136, Q138]*
- ⚠️ Trap: Ammonia mein N_2:H_2 mass ratio hamesha 14:3 hota hai — mole ratio (1:3) se confuse mat karna. *[Q110]*